Prime Decomposition

1. Express 24 as a product of its prime factors in index form.
2. Express 36 as a product of its prime factors in index form.
3. Express 60 as a product of its prime factors in index form.
4. Express 98 as a product of its prime factors in index form.
5. Express 140 as a product of its prime factors in index form.
6. Express 216 as a product of its prime factors in index form.
7. Express 360 as a product of its prime factors in index form.
8. Express 504 as a product of its prime factors in index form.
9. Given that $180 = 2^2 \times 3^2 \times 5$, use prime decomposition to find the smallest positive integer k such that $180k$ is a square number.
10. Given that $120 = 2^3 \times 3 \times 5$, use prime decomposition to find the smallest positive integer m such that $120m$ is a cube number.

Solutions

1. $2^3 \times 3$
   Explanation: Divide repeatedly by the smallest prime factors:

$$\begin{aligned} 24 &= 2 \times 12 \\ &= 2 \times 2 \times 6 \\ &= 2 \times 2 \times 2 \times 3 = 2^3 \times 3 \end{aligned}$$

2. $2^2 \times 3^2$
   Explanation: Break down 36 into prime factors:

$$\begin{aligned} 36 &= 4 \times 9 \\ &= (2 \times 2) \times (3 \times 3) = 2^2 \times 3^2 \end{aligned}$$

3. $2^2 \times 3 \times 5$
   Explanation: Factorise 60 step-by-step into primes:

$$\begin{aligned} 60 &= 6 \times 10 \\ &= (2 \times 3) \times (2 \times 5) = 2^2 \times 3 \times 5 \end{aligned}$$

4. 2×7^2
   Explanation: Divide 98 by 2 to obtain a square number:

$$\begin{aligned} 98 &= 2 \times 49 \\ &= 2 \times 7 \times 7 = 2 \times 7^2 \end{aligned}$$

5. $2^2 \times 5 \times 7$
   Explanation: Split 140 into convenient factors and reduce to primes:

$$\begin{aligned} 140 &= 14 \times 10 \\ &= (2 \times 7) \times (2 \times 5) = 2^2 \times 5 \times 7 \end{aligned}$$

6. $2^3 \times 3^3$
   Explanation: Divide by 2 repeatedly until an odd number remains, then divide by 3:

$$\begin{aligned} 216 &= 2 \times 108 \\ &= 2^2 \times 54 \\ &= 2^3 \times 27 = 2^3 \times 3^3 \end{aligned}$$

7. $2^3 \times 3^2 \times 5$
   Explanation: Factorise 360 using known decompositions ($360 = 36 \times 10$):

$$\begin{aligned} 360 &= (2^2 \times 3^2) \times (2 \times 5) \\ &= 2^3 \times 3^2 \times 5 \end{aligned}$$

8. $2^3 \times 3^2 \times 7$
   Explanation: Divide 504 repeatedly by small primes:

$$\begin{aligned} 504 &= 2 \times 252 \\ &= 2^2 \times 126 \\ &= 2^3 \times 63 \\ &= 2^3 \times 9 \times 7 = 2^3 \times 3^2 \times 7 \end{aligned}$$

9. $k = 5$
   Explanation: For a positive integer to be a square number, every exponent in its prime decomposition must be even. In $180 = 2^2 \times 3^2 \times 5^1$, the prime 5 has an odd exponent of 1. Multiplying by $k = 5$ gives:

$$180 \times 5 = 2^2 \times 3^2 \times 5^2 = (2 \times 3 \times 5)^2 = 30^2 = 900$$

   Thus, the smallest such integer is $k = 5$.

10. $m = 225$
    Explanation: For a positive integer to be a cube number, every exponent in its prime decomposition must be a multiple of 3. In $120 = 2^3 \times 3^1 \times 5^1$:

    (a) The exponent of 2 is already 3.

    (b) The exponent of 3 is 1, so we need 3^2 to reach 3^3.

    (c) The exponent of 5 is 1, so we need 5^2 to reach 5^3.

    Multiplying by $m = 3^2 \times 5^2 = 9 \times 25 = 225$ gives $120 \times 225 = 2^3 \times 3^3 \times 5^3 = 30^3 = 27{,}000$.